No.	Status	Date
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UT Re-measurement - 6-RA-103-11

Scope and Responsibilities

This report details the ultrasonic thickness re-measurement of 6-RA-103-11, a critical component within the ATU-100. The inspection was performed to validate the current wall thickness following the trim restoration of the upstream let-down valve. Sarah Thompson, acting as the UT technician, conducted the field measurements and data acquisition. David Park provided technical oversight and quality assurance for the inspection process.

The primary objective was to assess the local metal loss against the retirement thickness of 4.57 mm. All activities were executed in accordance with QSCP-NDT-001 Rev 4, which governs site NDE procedures for rich-amine fittings. The measurement techniques utilized for this assessment were performed under the guidelines established in ASTM E797 to ensure high-resolution data for the 10.97 mm nominal wall thickness piping.

Inspection Methodology and References

The inspection interval and specific CML selection were dictated by QSCP-IP-ATU100 Rev 4. This document outlines the risk-based inspection strategy for the Amine Treating Unit, specifically requiring a 12-month re-measurement for 6-RA-103-11 in February 2026. The data collected during this mobilization serves to update the short-term corrosion rate calculations required by site rules.

Sarah Thompson utilized calibrated ultrasonic equipment to perform manual thickness measurements across the identified area of thinning. The procedure QSCP-NDT-001 Rev 4 was strictly followed to ensure CML-id validation during the data export process. David Park reviewed the resulting thickness profiles to confirm compliance with the Larkhall NDE Services Inc. quality management system and the applicable NDT standards.

Assessment of Local Metal Loss

The current thickness readings for 6-RA-103-11 indicate that the component remains above the t-min of 4.57 mm. The observed thinning is progressing at approximately the line rate established since the let-down valve trim restoration. These findings are consistent with the projections provided in CIE-FFS-2025-005-R0, which performed a Level 2 Fitness-for-Service assessment on the local metal loss.

CIE-FFS-2025-005-R0 assumed the restoration of the let-down valve trim to mitigate accelerated erosion-corrosion. The current data confirms that the degradation remains within the acceptable limits defined by the FFS. Because the readings are above the retirement thickness, no non-conformance report was issued during this inspection cycle at the Hamilton Plant.

Conclusion and Future Requirements

Based on the 12-month re-measurement, 6-RA-103-11 is fit for continued service under the design conditions of 862 kPa(g) and 82 °C. The remaining life calculation, derived from the current thickness and the short-term corrosion rate, supports the existing inspection frequency. The component continues to meet the safety requirements for the rich amine header service.

Future inspections must adhere to the schedule defined in QSCP-IP-ATU100 Rev 4. The next thickness measurement for this CML should be scheduled at no more than half the remaining life, ensuring it does not exceed the class maximum for Class 2 amine service. Sarah Thompson and David Park have verified that the data meets all site-specific requirements for rich-amine piping integrity.

Thickness readings, measured 2026-02-16

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-11	10.97	4.57	2024-12-10	5.82	5.53	0.400	0.534	1.8	2027-02-16	gauge photo attached

Sign-off	Name	Function	Date	Signature
Prepared	Sarah Thompson	UT technician, SNT-TC-1A UT Level II	2026-02-16	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2026-02-16	_____